

Notes: Bornstein, Krusell & Rebelo (RES, 2023)

A World Equilibrium Model of the Oil Market

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1 Big picture

- **Question.** How should we model the world oil market when oil supply is produced by heterogeneous technologies, some firms behave competitively, and OPEC behaves strategically as a cartel?
- **Core innovation.** The paper builds a structural general-equilibrium model of the world economy with a detailed oil sector. The oil sector has OPEC conventional producers, non-OPEC conventional producers, and non-OPEC fracking producers.
- **Data advantage.** The authors use Rystad Energy micro data on roughly 14,000 oil fields, 3,200 companies, and 109 countries. These data identify technological differences between conventional oil and fracking.
- **Two fracking facts.** Fracking has a much shorter investment-to-production lag and much lower short-run adjustment costs than conventional production. The estimated average lag is about one year for fracking and about twelve years for conventional oil.
- **Market-structure assumption.** OPEC firms act as a cartel and internalize their collective market power. Non-OPEC firms are a competitive fringe.
- **Main quantitative result.** Fracking lowers the long-run real oil price by roughly 46 percent and raises world output by roughly 5 percent.
- **Volatility result.** Fracking lowers long-run oil price volatility because fracking producers respond quickly to demand and supply shocks. At the same time, it can raise world GDP volatility because a more elastic oil supply amplifies productivity shocks.
- **OPEC result.** Fracking weakens OPEC's market power, but OPEC does not necessarily lose much market share. It can preserve share by expanding production. Much of fracking's rise crowds out conventional non-OPEC production.
- **Big takeaway.** The oil market cannot be understood with either a pure Hotelling model or a purely competitive supply curve. The interaction of exploration lags, extraction-cost convexity, fracking flexibility, and OPEC strategic behavior is central.

2 Incremental contribution of the paper

- **Relative to commodity-cycle models.** Classic cobweb or cattle-cycle models often explain price-investment comovement using backward-looking expectations. This paper shows that a high correlation between oil prices and oil investment can arise from rational forward-looking investment when demand shocks are persistent and production has long lags.
- **Relative to Hotelling-style exhaustible-resource models.** The paper embeds oil exploration and reserve discovery into a setting with a steady state. This explains why the long-run real price of oil need not trend upward even though oil is a finite resource in a narrow physical sense.
- **Relative to SVAR work on oil shocks.** Structural VARs estimate demand and supply shocks in stationary environments. This paper provides a structural model that can study a large technological transition: the arrival and expansion of fracking.
- **Relative to micro studies of fracking.** Existing micro studies document features of fracking or local effects. This paper uses micro evidence to discipline a world general-equilibrium model and evaluates the global consequences of fracking.
- **Relative to models with competitive oil supply.** The paper explicitly models OPEC as a cartel facing a competitive fringe. This is important because fracking changes not only total supply but also the strategic environment faced by OPEC.
- **Methodological contribution.** The paper combines direct micro estimates of technology parameters with GMM estimation of aggregate parameters. It then uses identification diagnostics to show which moments discipline demand elasticity, shock persistence, and shock volatility.

- **Policy contribution.** The results speak to energy security and climate-transition debates: a more flexible fossil-fuel technology can stabilize prices but also change the macroeconomic transmission of productivity and supply shocks.

3 Data and measurement

3.1 Oil-field micro data

- The core data come from Rystad Energy and cover 1970–2019.
- The data include annual production, reserves, operational costs, and investment for oil fields around the world.
- Investment is the sum of exploration expenditures and capital expenditures.
- Production costs include lease costs plus fixed and variable production costs.
- Proven reserves are measured using a consistent economic and technical feasibility criterion.

Interpretation. The micro data matter because the key fracking mechanism is technological. The model needs direct evidence on production lags and adjustment costs, not just aggregate prices and quantities.

3.2 Three producer groups

The paper separates oil producers into:

- OPEC conventional producers;
- non-OPEC conventional producers;
- non-OPEC fracking producers.

Interpretation. This classification is the empirical counterpart of the model’s market structure. OPEC has strategic market power, while non-OPEC conventional and fracking firms form competitive fringes with different technologies.

3.3 Micro estimates of production lags

For each oil field, the paper measures the lag between first investment and first production. If production arrives according to a Poisson process, then the lag follows a geometric distribution and

$$\hat{\lambda} = \frac{N}{\sum_{i=1}^N T_i}, \quad (1)$$

where T_i is the lag for field i .

Interpretation. The arrival rate is high for fracking and low for conventional oil. This simple micro estimate becomes a central structural parameter in the general-equilibrium model.

3.4 Micro estimates of extraction adjustment costs

The paper estimates the elasticity of operational costs with respect to extraction rates. The regression logic is that if costs are strongly convex in the extraction rate, then producers cannot cheaply raise production from existing reserves.

Interpretation. The fracking estimate is much lower than the conventional-oil estimate. This is the second micro fact that makes fracking a flexible supply technology.

3.5 Aggregate moments

The model is disciplined with aggregate oil moments, including:

- volatility of oil prices;
- volatility of OPEC and non-OPEC investment;
- volatility of OPEC and non-OPEC production;
- correlations between prices, investment, and production;
- serial correlations of oil-price and oil-production changes.

Interpretation. The model has few parameters relative to many moments, so the paper is not simply fitting each outcome with its own free parameter. Identification comes from the joint behavior of prices, investment, and production across OPEC and non-OPEC producers.

4 Model and key equations

4.1 Representative household

The household owns oil firms, supplies labor, accumulates capital, and consumes the final good:

$$\max_{\{C_t, K_{t+1}, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{1-\gamma} \left(C_t - \varphi \frac{N_t^{1+\nu}}{1+\nu} \right)^{1-\gamma} \quad (2)$$

subject to

$$C_t + K_{t+1} \leq (1 + r_t - \delta)K_t + w_t N_t + \Pi_t^O. \quad (3)$$

Interpretation. Oil profits feed back into household income and aggregate demand. This is what makes the oil industry part of a world general-equilibrium model rather than a partial-equilibrium supply curve.

4.2 Final goods and oil demand

Final output is produced with value added and oil in a CES aggregator:

$$Y_t = \left[(1 - s_o)(A_t K_t^\alpha N_t^{1-\alpha})^{\frac{\epsilon-1}{\epsilon}} + s_o O_t^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}. \quad (4)$$

The first-order condition for oil gives a demand curve that can be written approximately as

$$\ln p_t = \ln s_o + \frac{1}{\epsilon} \ln Y_t - \frac{1}{\epsilon} \ln O_t. \quad (5)$$

Interpretation. The estimated oil demand elasticity is low. A small change in production can generate a large price movement. This is why oil-price volatility can remain high even when quantity movements are modest.

4.3 Non-OPEC conventional producers

A competitive non-OPEC conventional producer chooses investment and extraction. Reserves evolve as

$$R_{t+1}^N = (1 - \theta_t^N) R_t^N + \lambda X_{t+1}^N, \quad (6)$$

where θ_t^N is the extraction rate, R_t^N is reserves, X_t^N is exploration capital, and λ governs the lag between investment and production.

Exploration capital evolves as

$$X_{t+1}^N = (1 - \lambda) X_t^N + (I_t^N)^\xi (L^N)^{1-\xi}. \quad (7)$$

Interpretation. Investment does not instantly become production. This lag is the key reason investment can rise in response to high oil prices while reserves and production respond only gradually.

4.4 Fracking producers

Fracking producers have the same general structure as non-OPEC conventional firms but different technology parameters:

$$\lambda^F > \lambda^N, \quad \eta^F < \eta^N. \quad (8)$$

Interpretation. Fracking is faster and more flexible. It can respond to shocks in a way that conventional production cannot.

4.5 OPEC cartel problem

OPEC chooses extraction and investment strategically. Its objective can be written as

$$E_0 \sum_{t=0}^{\infty} \beta^t M_t [p_t e^{u_t} \theta_t^O R_t^O - I_t^O - \psi(e^{u_t} \theta_t^O R_t^O)^\eta], \quad (9)$$

subject to reserve dynamics, exploration-capital dynamics, the final-goods equilibrium conditions, and the optimality conditions of the competitive fringe.

Interpretation. OPEC is not a price taker. Its production decisions account for how the fringe reacts, how oil demand responds, and how prices affect world output.

4.6 Oil market clearing

The oil market clears according to

$$e^{u_t} \theta_t^O R_t^O + \theta_t^N R_t^N + \theta_t^F R_t^F = O_t. \quad (10)$$

Interpretation. Fracking enters the same market as conventional oil. Its macroeconomic effect comes from shifting the strategic and technological structure of world oil supply.

4.7 Hotelling rule as a limiting case

The classic Hotelling rule emerges when there is no OPEC, no reserve discovery, linear extraction costs, and risk-neutral households:

$$E_t(p_{t+1} - \psi) = \beta E_t(p_{t+2} - \psi). \quad (11)$$

In the full model with new reserve discovery and convex extraction costs, a steady state with constant real oil prices can exist.

Interpretation. The paper preserves the logic of intertemporal resource extraction but modifies it with exploration, market power, and convex adjustment costs. This helps explain why real oil prices do not show a strong long-run upward trend.

5 Identification and empirical design

5.1 What is directly estimated from micro data

- The production-arrival parameter λ is estimated from field-level investment-to-production lags.
- The extraction-cost convexity parameter η is estimated from field-level cost and extraction regressions.
- These estimates are separately informative for conventional oil and fracking.

Interpretation. This prevents the model from using aggregate moments alone to infer fracking flexibility. The key technology difference is tied to observed field-level behavior.

5.2 What is estimated by GMM

The remaining aggregate parameters are estimated by matching oil-price, investment, and production moments. In stylized notation:

$$\hat{\Theta} = \arg \min_{\Theta} [m^{data} - m^{model}(\Theta)]' W [m^{data} - m^{model}(\Theta)]. \quad (12)$$

Interpretation. The parameters are not chosen to fit the fracking transition directly. They are estimated from pre-fracking and aggregate oil-market dynamics, then used to evaluate fracking.

5.3 Why identification is credible

- Oil demand elasticity is identified by the volatility of prices relative to output and production.
- Demand shock persistence is identified by the high correlation between oil prices and oil investment.
- Supply shock volatility is identified partly by the relative volatility and serial correlation of OPEC production.
- The split between demand and supply shocks is identified by the different behavior of OPEC and non-OPEC production.

Interpretation. The identification logic is economic, not only statistical. The model assigns specific roles to price volatility, investment persistence, and OPEC/non-OPEC production patterns.

5.4 Potential limitations

- OPEC is treated as a cartel with a particular commitment structure.
- Fracking forecasts rely partly on Rystad projections for future market shares.
- The model is solved by linearization around a steady state, which may understate large nonlinear transitions.
- The model abstracts from climate policy, geopolitical sanctions, financial constraints, and heterogeneous demand sectors.

Interpretation. The paper’s strength is structural discipline, but this comes with assumptions. The results are best read as model-implied consequences of fracking under a specific, economically motivated structure.

6 Tables and figures

6.1 Table 1 and Figure 1: descriptive facts and production lags

Read. Table 1 shows that OPEC’s market share is stable at about 42 percent across the two sample periods, while fracking rises from essentially zero to 5 percent. Non-OPEC conventional producers lose share as fracking expands. Figure 1 shows that conventional fields have a long right tail of investment-to-production lags, while fracking fields move into production quickly.

Interpretation. The table and figure establish the main empirical puzzle and the main technological difference. Fracking grows without immediately destroying OPEC’s share; instead, the burden of adjustment falls largely on conventional non-OPEC producers. The lag evidence explains why fracking can be a much more shock-responsive technology.

Economic message. This is the empirical foundation for the paper’s model: a world oil market with heterogeneous technologies and strategic OPEC behavior.

TABLE 1
Descriptive statistics

Time period	1970–2010			2011–19		
Firms	OPEC conv.	Non-OPEC conv.	Non-OPEC fracking	OPEC conv.	Non-OPEC conv.	Non-OPEC fracking
Average market share	42%	58%	0%	42%	53%	5%
Average production (Mbbl)	9,227	12,548	41	11,330	14,132	1,454
Average no. of operating oil fields	799	5,653	139	1,220	9,211	1,086

Notes: This table presents descriptive statistics of the three groups of firms in the data: OPEC conventional producers, non-OPEC conventional producers, and non-OPEC fracking producers. We split the sample into two subperiods: 1970–2010 and 2011–19. In 2011, fracking producers surpassed for the first time a 1% market share in world oil production. Mbbl, million barrels of oil.

(a) Table 1: descriptive statistics

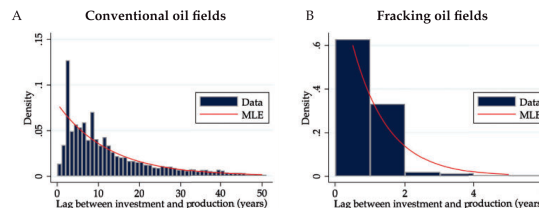


FIGURE 1

(b) Figure 1: investment-to-production lags

6.2 Table 2 and Figure 2: adjustment costs and fracking responsiveness

Read. Table 2 estimates extraction-cost convexity. Conventional fields have much larger adjustment-cost coefficients than fracking fields. Figure 2 shows a close high-frequency relation between US oil rigs and oil prices, especially in the fracking era.

Interpretation. The coefficient for fracking is far below the conventional estimates, which means that raising extraction is cheaper for fracking firms. The rig-count evidence gives an intuitive time-series counterpart: rigs move strongly with prices, but also fall sharply when prices collapse.

Economic message. Fracking is not just more oil supply; it is more flexible supply. That flexibility is what changes price volatility and OPEC’s residual demand curve.

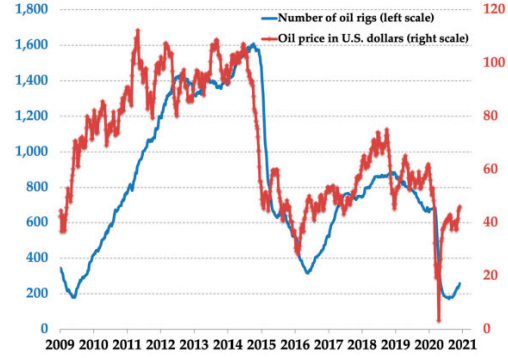
TABLE 2
Extraction rate adjustment costs regression

Dep. variable: ln(prod. costs per barrel of oil reserves)

Variable	(1)	(2)	(3)	(4)
ln(extraction)	6.04*** (0.34)	6.11*** (0.37)	2.23*** (0.75)	6.79*** (1.21)
Oil field FE	✓	✓	✓	✓
Operation year FE	✓	✓	✓	✓
Sample	All	Non-OPEC	Non-OPEC	OPEC
Technology	All	Conventional	Fracking	Conventional
IV	✓	✓	✓	✓
1st stage F-stat	228	194	3	24
Clusters (oil fields)	13,506	11,103	939	1,464
Observations	303,457	260,552	8,819	36,086

Notes: This table presents the regression results for the adjustment cost coefficient, η . Standard errors are clustered at the oil field level. The instrument used is the 1-year-ahead forecast of detrended world real GDP. *** significant at the 1% level.

(a) Table 2: extraction-rate adjustment costs



(b) Figure 2: rigs and oil prices

6.3 Table 3: investment-price correlation for top producers

Read. For most large oil companies, investment growth is positively correlated with oil-price growth. This pattern holds across OPEC and non-OPEC firms, although the correlations vary by firm.

Interpretation. Positive price-investment correlations could be misread as backward-looking overinvestment. The paper interprets them differently: persistent demand shocks make high prices informative about future profitability, so forward-looking firms rationally invest when prices rise.

Economic message. The table motivates the need for a dynamic model with investment lags and persistent demand shocks.

TABLE 3
Investment and price correlation for top 20 firms

Firm	Headquarters	OPEC	corr($\Delta p_t, \Delta i_t$)
Saudi Aramco	Saudi Arabia	✓	0.31
Rosneft	Russia	✗	0.34
PetroChina	China	✗	0.36
Kuwait Petroleum Corp (KPC)	Kuwait	✓	0.3
NIOC (Iran)	Iran	✓	0.06
Pemex	Mexico	✗	0.27
ExxonMobil	US	✗	0.35
Lukoil	Russia	✗	0.41
Petrobras	Brazil	✗	0.3
PDVSA	Venezuela	✓	0.28
Abu Dhabi NOC	Abu Dhabi	✓	0.14
Chevron	US	✗	0.43
Shell	Netherlands	✗	0.34
BP	UK	✗	0.35
Surgutneftegas	Russia	✗	0.26
South Oil Company (Iraq NOC)	Iraq	✓	0.19
Total	France	✗	0.04
CNOOC	China	✗	0.4
Statoil	Norway	✗	0.34
Eni	Italy	✗	0.04

Notes: This table presents the correlation between the growth rate of investment and the growth rate in oil prices for the top-20 oil producers, in descending order of production volume. In the table, x_t represents the logarithm of X_t , Δx_t is equal to $x_t - x_{t-1}$, and p_t and i_t represent the logarithm of the real price of oil and the logarithm of the firm's real

Table 3: investment and price correlation for top 20 producers

6.4 Tables 4 and 5: model fit and estimated parameters

Read. Table 4 compares twenty data moments with model-implied moments. The model matches many volatilities and correlations reasonably well with only a small number of structural parameters. Table 5 reports the estimated oil demand elasticity $\epsilon = 0.15$, persistent demand-shock parameters, and shock volatilities.

Interpretation. The low demand elasticity is central: a one percent increase in oil production lowers the oil price by roughly seven percent. The model's fit is imperfect, but it is disciplined by a demanding set of

moments rather than by a small number of handpicked targets.

Economic message. The model is parsimonious but powerful. It has enough structure to connect micro oil technology to macro oil-price dynamics.

TABLE 4
Data and model moments

	Moment	Data	(s.e.)	Model
(1)	$\text{std}(\Delta p_t)$	0.27	(0.03)	0.21
(2)	$\text{std}(\Delta i_t^N)$	0.18	(0.03)	0.16
(3)	$\text{std}(\Delta i_t^D)$	0.17	(0.03)	0.21
(4)	$\text{std}(\Delta q_t^N)$	0.02	(0.003)	0.02
(5)	$\text{std}(\Delta q_t^D)$	0.07	(0.01)	0.08
(6)	$\text{corr}(\Delta p_t, \Delta i_t^N)$	0.49	(0.14)	0.61
(7)	$\text{corr}(\Delta p_t, \Delta i_t^D)$	0.33	(0.11)	0.60
(8)	$\text{corr}(\Delta p_t, \Delta q_t^N)$	0.00	(0.09)	0.08
(9)	$\text{corr}(\Delta p_t, \Delta q_t^D)$	0.04	(0.14)	-0.68
(10)	$\text{corr}(\Delta i_t^N, \Delta i_t^D)$	0.67	(0.12)	1.00
(11)	$\text{corr}(\Delta i_t^N, \Delta q_t^N)$	0.07	(0.12)	0.10
(12)	$\text{corr}(\Delta i_t^N, \Delta q_t^D)$	0.04	(0.13)	0.04
(13)	$\text{corr}(\Delta i_t^D, \Delta q_t^N)$	-0.11	(0.18)	0.08
(14)	$\text{corr}(\Delta i_t^D, \Delta q_t^D)$	-0.28	(0.13)	0.04
(15)	$\text{corr}(\Delta q_t^N, \Delta q_t^D)$	-0.07	(0.14)	0.40
(16)	$\text{corr}(\Delta p_t, \Delta p_{t-1})$	-0.06	(0.09)	-0.31
(17)	$\text{corr}(\Delta i_t^N, \Delta i_{t-1}^N)$	0.13	(0.16)	0.05
(18)	$\text{corr}(\Delta i_t^D, \Delta i_{t-1}^D)$	0.29	(0.11)	0.04
(19)	$\text{corr}(\Delta q_t^N, \Delta q_{t-1}^N)$	0.67	(0.10)	0.71
(20)	$\text{corr}(\Delta q_t^D, \Delta q_{t-1}^D)$	0.25	(0.20)	-0.31

Notes: This table presents the targeted moments from the data and the model-implied moments under the benchmark specification. Newey-West standard errors computed with 5-year lags in parentheses. In the table, x_t represents the logarithm of X_t , Δx_t is equal to $x_t - x_{t-1}$, and the variables q_t^N and q_t^D represent the logarithm of oil production for non-OPEC and OPEC firms, respectively.

(a) Table 4: data and model moments

TABLE 5
Estimated parameters

Parameter	Estimate	(s.e.)
ϵ	0.15	(0.003)
ρ_1^A	1.72	(0.07)
ρ_2^A	-0.74	(0.08)
$\text{std}(\epsilon_t^A)$	0.013	(0.002)
$\text{std}(u_t)$	0.05	(0.005)

Notes: This table presents the benchmark GMM estimates of the structural parameters.

(b) Table 5: estimated parameters

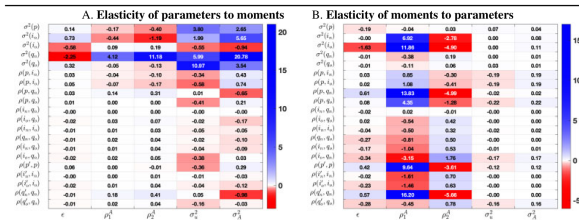
6.5 Tables 6 and 7: sources of identification

Read. Table 6 shows how estimated parameters respond to individual moments and how moments respond to parameters. Table 7 transforms shock parameters into total shock volatility and the relative importance of demand shocks.

Interpretation. The heat maps are useful because they reveal the economics behind identification. Demand elasticity is tied to price volatility and production/investment volatility. The demand-versus-supply split is disciplined by how OPEC production volatility differs from non-OPEC production volatility.

Economic message. The model is not a black box. The tables show why particular moments push particular parameters and why simply raising shock volatility cannot solve all fit problems.

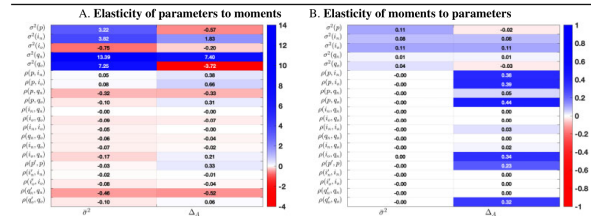
TABLE 6
Sources of identification



Notes: The left panel presents the elasticity of estimated parameters to the targeted moments, following the approach of

(a) Table 6: sources of identification

TABLE 7
Sources of identification—demand vs. supply shocks



(b) Table 7: demand versus supply shocks

6.6 Table 8 and Figure 3: short-run elasticity and the role of investment lags

Read. Table 8 estimates the short-run elasticity of extraction rates with respect to oil prices. The IV estimates are about 0.20–0.21 for all producers, non-OPEC producers, and OPEC producers. Figure 3 compares impulse responses when the investment-production lag is long versus when there is essentially no lag.

Interpretation. Table 8 shows that existing oil fields respond only modestly to price changes in the short run. Figure 3 shows why the lag matters: if investment instantly creates reserves, reserves rise after a demand shock; with a long lag, reserves can fall while investment is still building future supply.

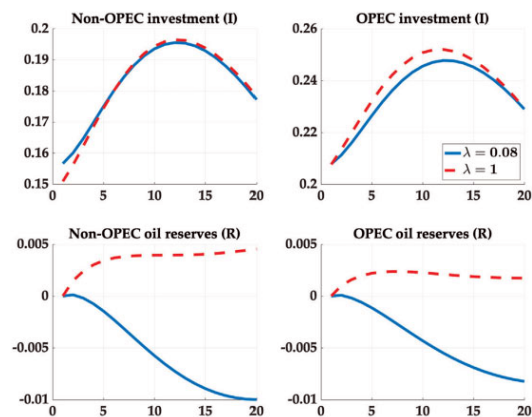
Economic message. Short-run supply is inelastic not because firms are irrational, but because production technology and investment lags constrain them.

TABLE 8
Price elasticity of extraction rates

Dep. variable: $\ln(\text{extraction rate})$	(1)	(2)	(3)	(4)
Variable	(1)	(2)	(3)	(4)
$\ln(\text{price})$	0.05*** (0.005)	0.20*** (0.009)	0.21*** (0.010)	0.21*** (0.027)
IV	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
Sample	All	All	Non-OPEC	OPEC
Clusters (oil fields)	11,553	10,958	9,471	1,217
Observations	229,985	229,390	201,386	28,004

Notes: This table presents estimates of the elasticity of extraction rates with respect to the price of oil. The data used include all oil fields with available extraction rates in 1970–2010, excluding the first year of operation. The estimation

(a) Table 8: price elasticity of extraction rates



(b) Figure 3: impulse responses for different λ

6.7 Tables 9–11: reserves, costs, and long-run fracking effects

Read. Table 9 shows that a no-lag model produces counterfactual correlations between oil-price changes and reserve changes. Table 10 shows that convex extraction costs are needed to generate volatile average production costs. Table 11 reports the headline long-run fracking effects: oil prices fall 46 percent, world output rises 5 percent, oil-price volatility falls, and OPEC’s market share falls only modestly.

Interpretation. These tables clarify why the model has both lagged investment and convex extraction costs. Without lags, reserve dynamics are wrong; without convexity, production-cost dynamics are wrong. Table 11 then uses the calibrated fracking technology to quantify the long-run equilibrium change.

Economic message. Fracking lowers prices partly by expanding supply and partly by weakening OPEC’s market power. The output gain is much larger when OPEC is strategic than it would be in a purely competitive model.

TABLE 9
Correlation between oil prices and reserves

	$\text{corr}(\Delta p, \Delta R_o)$	$\text{corr}(\Delta p, \Delta R_n)$
Data	-0.05	-0.02
(s.e.)	(0.17)	(0.14)
Benchm. model ($\lambda = 0.08$)	0.10	-0.12
No-lag model ($\lambda = 1$)	0.68	0.38

Notes: This table presents the correlation between oil price changes and oil reserve changes for OPEC and non-OPEC producers in the data, in the benchmark model, and in the model with no lag between investment and production.

	$\text{std}(\Delta AC_o)$	$\text{std}(\Delta AC_n)$
Data	0.24	0.21
(s.e.)	(0.06)	(0.06)
Benchm. model ($\eta = 6.11$)	0.4	0.12
Linear production ($\eta = 0$)	0.0	0.0

Notes: This table presents the volatility of average production cost changes for OPEC and non-OPEC producers in the data, in the benchmark model, and in the model with linear production costs.

TABLE 11
Implications of fracking for key aggregate moments

Moment	No fracking	Fracking
Average p_t		-46%
Average y_t		+5%
Volatility of Δp_t	0.21	0.12
Volatility of Δy_t	0.035	0.036
OPEC's market share	42.3%	40.1%

Notes: This table presents the aggregate long-run impact of the advent of fracking. In the table, x_t represents the

(a) Table 9: price-reserve correlations

(b) Table 10: average-cost volatility

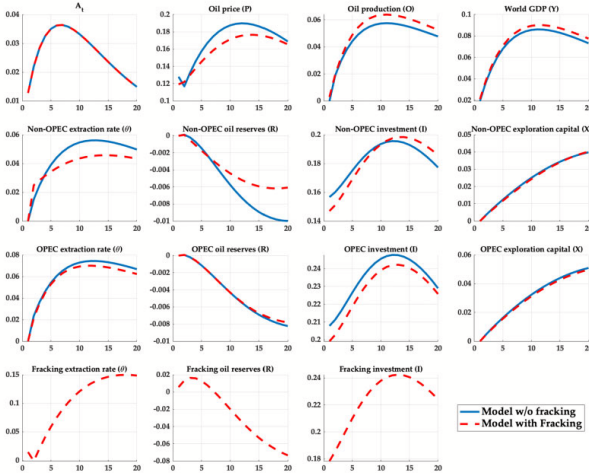
(c) Table 11: long-run effects of fracking

6.8 Figures 4 and 5: demand shocks versus supply shocks

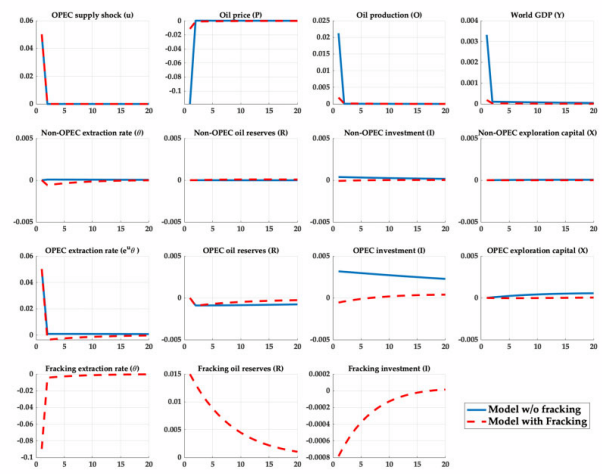
Read. Figure 4 shows that a demand shock raises oil prices, oil production, investment, and world GDP. With fracking, oil supply is more elastic, so oil prices respond less and quantities respond more. Figure 5 shows that an OPEC supply shock has a much smaller effect on oil prices when fracking is present.

Interpretation. The figures reveal a subtle macro result. Fracking stabilizes oil prices after supply disturbances because fracking can quickly offset OPEC disruptions. But after productivity or demand shocks, the same elasticity lets the real economy expand more, raising GDP volatility.

Economic message. Supply flexibility is not the same as macro stability. It stabilizes oil prices but can amplify non-oil shocks by making energy less of a bottleneck.



(a) Figure 4: demand shock



(b) Figure 5: supply shock

6.9 Table 12 and Figures 6–7: variance decomposition and fracking transition

Read. Table 12 shows that, without fracking, supply shocks explain most oil-price volatility. With fracking, oil-price volatility becomes overwhelmingly demand-driven. Figure 6 compares data and model forecasts for fracking's market share. Figure 7 shows transition dynamics: oil production rises, oil prices fall sharply, world GDP rises, and fracking crowds out conventional non-OPEC supply more than OPEC supply.

Interpretation. Table 12 is one of the clearest pieces of evidence that fracking changes the nature of oil-market fluctuations. Figure 6 checks that the model's fracking transition is not wildly inconsistent with observed and forecast market shares. Figure 7 shows why OPEC remains relevant: it expands production enough to protect much of its share, while conventional non-OPEC producers lose ground.

Economic message. Fracking is a structural change, not just a short-run shock. It changes long-run price levels, volatility sources, and market-share composition.

TABLE 12
Variance decomposition

Moment	No fracking		Fracking	
	Demand (%)	Supply (%)	Demand (%)	Supply (%)
Δp_t	39.0	61.0	98.4	1.6
Δi_t	99.9	0.01	99.99	0.01
Δq_t	42.8	57.2	99.1	0.9

Notes: This table presents the variance decomposition of five key variables to demand and supply shocks in the model without fracking and the one with fracking. In the table, x_t represents the logarithm of X_t , Δx_t is equal

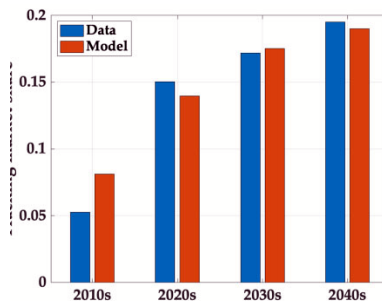


FIGURE 6

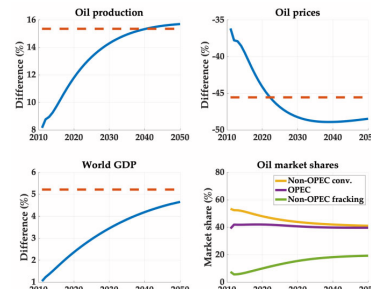


FIGURE 7

(a) Table 12: variance decomposition

(b) Figure 6: fracking share

(c) Figure 7: transition dynamics

7 Short summary

- The paper builds a general-equilibrium model of the oil market with OPEC, conventional non-OPEC producers, and fracking firms.
- The key micro facts are that fracking has much shorter production lags and much lower extraction adjustment costs than conventional oil.
- The model estimates oil demand elasticity at about 0.15, implying very inelastic demand.
- The model explains positive price-investment correlations through persistent demand shocks and forward-looking investment.
- Long lags between investment and production are essential for matching reserve dynamics.
- Convex production costs are essential for matching volatility in average production costs.
- Fracking lowers long-run oil prices by about 46 percent and raises world output by about 5 percent.
- Fracking lowers oil-price volatility because fracking firms can respond quickly to shocks.
- Fracking can raise world GDP volatility because oil supply becomes less of a dampening constraint on productivity shocks.
- OPEC's market share remains relatively stable because OPEC expands output, while fracking mostly crowds out conventional non-OPEC producers.

8 Conclusion

8.1 Core conclusion

Bornstein, Krusell, and Rebelo show that the world oil market is best understood as a general-equilibrium system with heterogeneous production technologies and strategic market power. The arrival of fracking is not merely an increase in reserves or a shift in a supply curve. It changes the speed at which supply can respond, the residual demand facing OPEC, and the transmission of aggregate shocks.

8.2 What the paper teaches about oil prices

The paper explains why oil prices can be volatile even when real oil prices do not trend upward in the long run. Demand is very inelastic, so small quantity changes produce large price movements. At the same time,

exploration can replenish reserves, which avoids the stark prediction of a simple Hotelling model that real scarcity rents must rise persistently. The result is a model in which oil prices fluctuate sharply around a long-run steady state.

8.3 What the paper teaches about fracking

Fracking has two distinct effects. First, it expands the oil supply frontier. Second, and more importantly, it changes the elasticity of supply at relevant horizons. Because fracking investment comes online quickly and extraction rates are easier to adjust, fracking firms can respond faster to price movements than conventional producers. This makes oil prices less sensitive to supply disruptions but makes the real economy more responsive to productivity shocks.

8.4 Implications for OPEC and market power

The model's OPEC result is subtle. Fracking weakens OPEC's cartel power, but OPEC does not necessarily lose much market share. Instead, it produces more to preserve strategic relevance. This means fracking can reduce prices and cartel rents even when OPEC's observed share looks stable. Market share alone is therefore not a sufficient measure of market power.

8.5 Policy and energy-transition implications

The paper has two policy-relevant implications. First, flexible supply technologies can reduce exposure to oil-price spikes caused by supply disruptions. Second, fossil-fuel flexibility can delay or complicate energy-transition dynamics by lowering prices and weakening the scarcity channel that would otherwise encourage substitution away from oil. A climate policy analysis would need to combine this oil-market structure with emissions externalities and policy constraints.

8.6 Research agenda

The framework opens several extensions. One could add financial constraints to oil exploration, heterogeneous demand sectors, strategic investment in spare capacity, climate policy, sanctions and geopolitical risk, or endogenous OPEC cohesion. The key methodological lesson is that micro oil-field data can discipline macro commodity models, making counterfactuals about structural change more credible.

8.7 Takeaway

The central takeaway is that fracking changed the global oil market by making supply faster and more elastic. This reduces oil-price volatility and weakens OPEC's market power, but it also changes macroeconomic shock transmission. The paper's incremental contribution is to quantify these mechanisms in a single model that is disciplined by both micro oil-field evidence and aggregate oil-market dynamics.